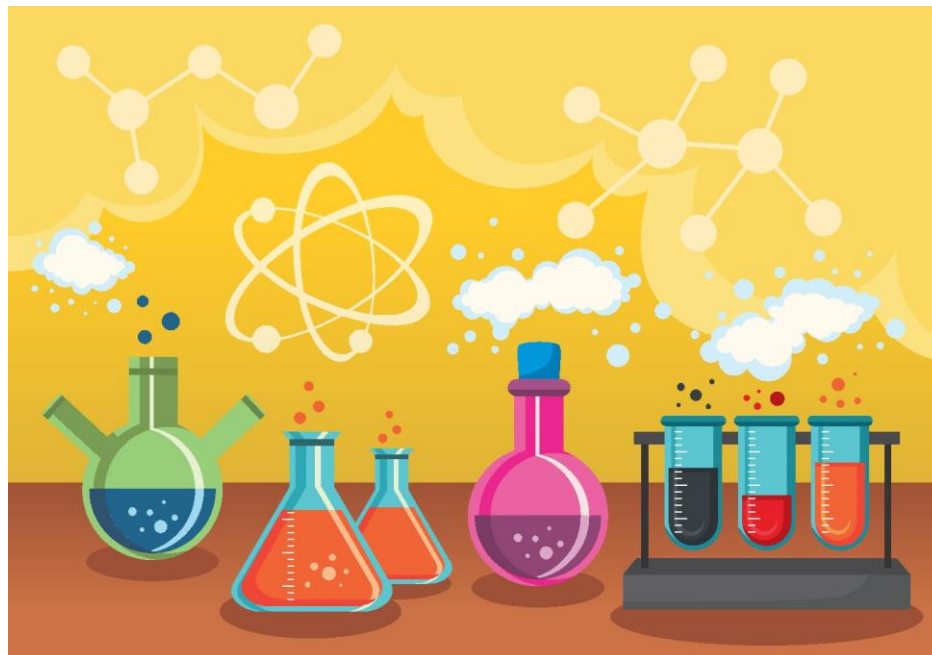


Student Name: _____

Class period: _____

AP Chemistry



Unit 2: Chemical Reactions

Textbook Reference: Zumdahl Chemistry: A Molecular Approach Chapter 1-4

Date	Day	Learning Targets	Activities	Work on this!
9/2-3	U2D1	What are physical and chemical changes?	- Review Exam 1 - Review phys/chem changes- self guided	Pink packet through pg 4
9/4-5	U2D2	How do we use stoichiometry?	Stoichiometry of rxns with masses, gases, and solutions	
9/8-9	U2D3	What is a limiting reactant?	- Limiting reactants - Return of the RICE table - RXN types	Pink packet through ~pg 14
9/10-11	U2D4	What is a net ionic equation?	POGIL: Net Ionic Equations	Finish POGIL Quiz next time
9/12-15	U2D5	How is stoichiometry used in lab procedures?	- Quiz - Common Lab Procedures: Gravimetric analysis and Titration - Lab: Titration	Lab due next time Finish pink packet
9/16-17	U2D6	What is gravimetric analysis?	Lab: Murder in the Chem Lab	Lab due next time? Test next time!
9/18-19	U2D7	What have you learned?	Unit 2 Exam	

Unit 2: Chemical Reactions

3.8	Representations of Solutions
3.8A	Using particulate models for mixtures: i. Represent interactions between components. ii. Represent concentrations of components.
	3.8.A.1: Particulate representations of solutions communicate the structure and properties of solutions, by illustration of the relative concentrations of the components in the solution and / or drawings that show interactions among the components.
4.1	Introduction to Chemical Reactions
4.1A	Identify evidence of chemical and physical changes in matter.
	4.1.A.1: A physical change occurs when a substance undergoes a change in properties but not a change in composition. Changes in the phase of a substance (solid, liquid, gas) or formation / separation of mixtures of substances are common physical changes.
	4.1.A.2: A chemical change occurs when substances are transformed into new substances, typically with different compositions. Production of heat or light, formation a gas, formation of a precipitate, and / or color change provide possible evidence that a chemical change has occurred.
4.2	Net Ionic Equations
4.2A	Represent changes in matter with a balanced chemical or net ionic equation: i. For physical changes. ii. For given information about the identity of the reactants and / or product. iii. For ions in a given chemical reaction.
	4.2.A.1: All physical and chemical processes can be represented symbolically by balanced equations.
	4.2.A.2: Chemical equations represent chemical changes. These changes are the result of a rearrangement of atoms into new combinations; thus, any representation of a chemical change must contain equal numbers of atoms of every element before and after the change occurred. Equations thus demonstrate that mass and charge are conserved in chemical reactions.
	4.2.A.3: Balanced molecular, complete ionic, and net ionic equations are differing symbolic forms used to represent a chemical reaction. The form used to represent the reaction depends on the context in which it is to be used.
4.3	Representations of Reactions
4.3A	Represent a given chemical reaction or physical process with a consistent particulate model.
	4.3.A.1: Balanced chemical equations in their various forms can be translated into symbolic particulate representations.
4.4	Chemical and Physical Changes
4.4A	Explain the relationship between macroscopic characteristics and bond interactions for: i. Chemical processes. ii. Physical processes.
	4.4.A.1: Processes that involve the breaking and / or formation of chemical bonds are typically classified as chemical processes. Processes that involve only changes in intermolecular interactions, such as phase changes, are typically classified as physical processes.
	4.4.A.2: Sometimes physical processes involve the breaking of chemical bonds. For example, plausible arguments could be made for the dissolution of a salt in water, as either a physical or chemical process, involves breaking of ionic bonds, and the formation of ion-dipole interactions between ions and solvent.
4.5	Stoichiometry
4.5A	Explain changes in the amounts of reactants and products based on the balanced reaction equation for a chemical process.
	4.5.A.1: Because atoms must be conserved during a chemical process, it is possible to calculate product amounts by using known reactant amounts, or to calculate reactant amounts given known product amounts.
	4.5.A.2: Coefficients of balanced chemical equations contain information regarding the proportionality of the amounts of substances involved in the reaction. These values can be used in chemical calculations involving the mole concept.
	4.5.A.3: Stoichiometric calculations can be combined with the ideal gas law and calculations involving molarity to quantitatively study gases and solutions.
4.7	Types of Chemical Reactions
4.7A	Identify a reaction as acid-base, oxidation-reduction, or precipitation.
	4.7.A.1: Acid-base reactions involve transfer of one or more protons (H ⁺ ions) between chemical species.
	4.7.A.2: Oxidation-reduction (redox) reactions involve transfer of one or more electrons between chemical species, as indicated by changes in oxidation numbers of the involved species. Combustion is an important subclass of oxidation-reduction reactions, in which a species reacts with oxygen gas. In the case of hydrocarbons, carbon dioxide and water are products of complete combustion.
	4.7.A.3: In a redox reaction, electrons are transferred from the species that is oxidized to the species that is reduced. ★
	4.7.A.4: Oxidation numbers may be assigned to each of the atoms in the reactants and products; this is often an effective way to identify the oxidized and reduced species in a redox reaction.
	4.7.A.5: Precipitation reactions frequently involve mixing ions in aqueous solution to produce an insoluble or sparingly soluble ionic compound. All sodium, potassium, ammonium, and nitrate salts are soluble in water. ★

Physical and Chemical Properties and Changes

Intensive & Extensive Properties

Extensive property: a characteristic that depends on the _____ of matter present (how much you have).

- | | | |
|---------|-----------|---------------------------|
| 1. Mass | 2. Volume | 3. Length (width, height) |
|---------|-----------|---------------------------|

Intensive property: a characteristic that does _____ depend on the amount of matter present.

- | | | | | |
|----------|----------------|-----------------|-----------------|-----------|
| 1. Color | 2. Density | 3. Malleability | 4. Ductility | 5. Luster |
| 6. Odor | 7. Melting Pt. | 8. Boiling Pt. | 9. Conductivity | |

Physical Properties/Changes

Physical property: a characteristic that can be observed or measured without changing the _____ of the substance.

- | | | | | |
|-------------------------------------|------------------|-----------------|--------------|------------|
| 1. Color | 2. Melting point | 3. Malleability | 4. Ductility | 5. Density |
| 6. Solubility (ability to dissolve) | 7. Mass | 8. Volume | 9. Viscosity | |

Physical change: a change when chemical bonds within compounds are _____ broken in the reactant and no new chemical bonds are formed in the products. The _____ of the material does NOT change.

1. Changes of _____. Example: $\text{Fe(s)} \rightarrow \text{Fe(l)}$ or $\text{CO}_2\text{(s)} \rightarrow \text{CO}_2\text{(g)}$
2. Changes in _____.

Chemical Properties/Changes

Chemical property: a characteristic that can ONLY be observed or measured by subjecting it to a process that might **change** the _____ of the substance.

- | | | | | |
|---|-----------------|-------------|-----------------------------------|-----------------------|
| 1. Reactivity | 2. Flammability | 3. Toxicity | 4. pH (how acidic or basic it is) | 5. Ability to Ferment |
| 6. Ability to Oxidize, for example: rusting (___) or tarnishing (___) | | | | |

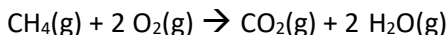
Chemical change: a change when chemical bonds within the compound ARE broken in the reactants and _____ chemical bonds are formed in the products. The _____ of the material changes.

Potential Indications of Chemical Change

1. Energy change (emission or absorption of heat or light)
2. Formation of a gas
3. Formation of a precipitate
4. Formation of an odor
5. Color change

Chemical Reactions can be described in several ways: you must be able to convert from any of these forms to another

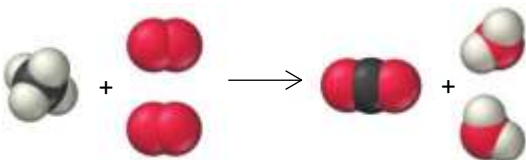
1. In a _____ (equation)



2. In a _____

Methane reacts with oxygen gas to form carbon dioxide and water vapor.

3. In a picture (_____ representation)



Symbols used in chemical equations:

————→ forms, produces, yields

(s) solid state

(l) liquid state

(g) gaseous state

(aq) aqueous solution (dissolved in water).

$\xrightarrow{\text{heat}}$ or $\xrightarrow{\Delta}$ Reactants are heated.

$\xrightarrow{\text{MnO}_2}$ Formula of catalyst used to speed up reaction. The catalyst does NOT undergo permanent chemical change itself.

Steps in a Chemical Reaction: Atoms are _____ created or destroyed! (Law of Conservation of _____)

1. Reactant bonds _____
2. Atoms rearranged.
3. _____ bonds formed in products!
4. Mass of all Reactants = Mass of all Products

Obeying the Law (of Conservation of Mass): Balancing Chemical Equations!

1. Word equations must be translated into a formula equation.

Do not forget nomenclature:

- Covalent compounds: prefixes
- Ionic compounds: balance charges
- Diatomic elements: **Br₂, I₂, N₂, Cl₂, H₂, O₂, F₂** ← But only when alone (i.e. not bonded to another element)

2. Balance 1 atom at a time by writing _____ in front of compound.

- Do NOT ever change subscripts!

3. Polyatomic ions treated as a single unit (_____ they stay together on both sides)

4. If an equation has C, H, and O's balance them in that respective order.

5. Make sure the coefficients are in the _____ whole number ratio (reduce).

Practice with Balancing Reactions!

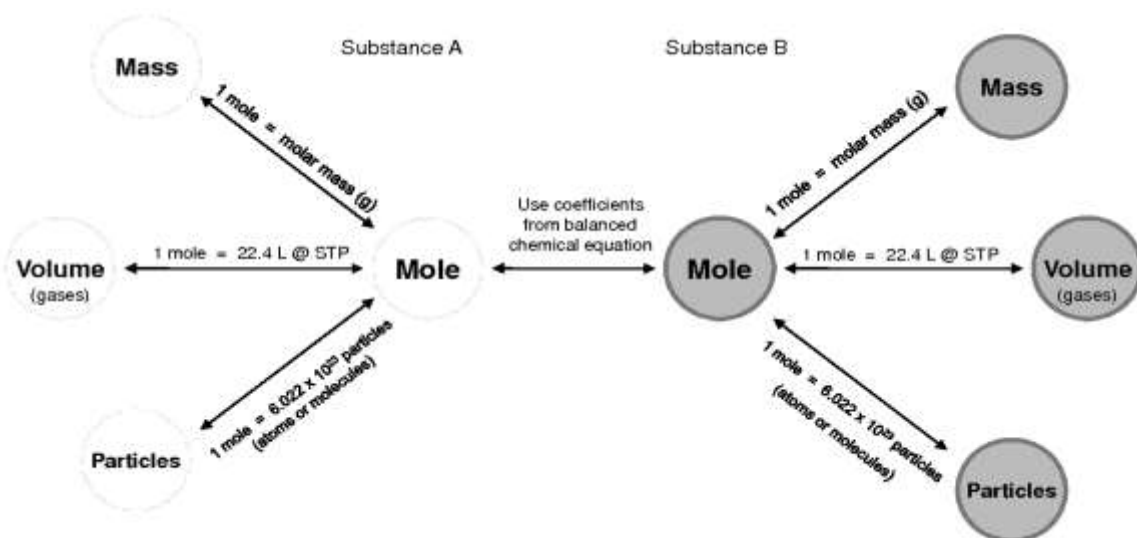
1. $\text{Fe(s)} + \text{O}_2\text{(g)} \rightarrow \text{Fe}_2\text{O}_3\text{(s)}$
2. $\text{C}_3\text{H}_8\text{(g)} + \text{O}_2\text{(g)} \rightarrow \text{H}_2\text{O(l)} + \text{CO}_2\text{(g)}$
3. $\text{C}_6\text{H}_{14}\text{(g)} + \text{O}_2\text{(g)} \rightarrow \text{H}_2\text{O(l)} + \text{CO}_2\text{(g)}$
4. Aluminum and molecular chlorine undergo a synthesis reaction. Write and balance the equation for this reaction.
5. An impure sample containing lead (II) nitrate and an inert material is added to sodium hydroxide. The two aqueous reactants undergo a chemical change, producing aqueous lead (II) hydroxide and one other compound dissolved in the solution. Write and balance the equation for this reaction.
6. An aqueous reaction of hydrochloric acid and potassium sulfite undergoes a gas evolution reaction to form liquid water, sulfur dioxide vapor, and aqueous potassium chloride. Write and balance the equation for this reaction.

Stoichiometry

Stoichiometry involves the study of the relationships between _____ and _____ in a chemical reaction.

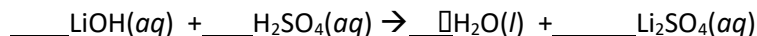
- The equation that is describing the reaction must be balanced.
- The coefficients in a chemical equation can describe individual _____ or amounts in _____ of the products and reactants.

Mole Ratios and Molar Mass as Conversion Factors: A _____ ratio is a conversion factor that relates the amounts in moles of any two chemical species involved in a chemical reaction.



Basic Stoichiometry: Gram/Mole/Particle Conversions

1. How many total moles of product are produced if 3.25 g of LiOH reacts with excess H_2SO_4 ?



2. Consider the following reaction: $\text{Ba}(\text{NO}_3)_2(aq) + 2 \text{KOH}(aq) \rightarrow \text{Ba}(\text{OH})_2(aq) + 2 \text{KNO}_3(aq)$

a. What mass of barium hydroxide could be formed from 1.0 kg of barium nitrate?

b. How many hydroxide ions would be in this sample?

Gas Stoichiometry: At STP and non-STP

3. 0.500 L of $\text{H}_2(g)$ reacts with excess $\text{O}_2(g)$ at STP according to the equation: $2 \text{H}_2(g) + \text{O}_2(g) \rightarrow 2 \text{H}_2\text{O}(g)$

What volume of water is produced?

4. How many grams of carbon react if 543 L of gaseous carbon monoxide can be produced at 27°C and 0.247 atm according to the following equation? $2 \text{C}(s) + \text{O}_2(g) \rightarrow 2 \text{CO}(g)$

5. Air bags in cars are inflated by the sudden decomposition of sodium azide, NaN_3 , by the following reaction: $2 \text{NaN}_3(s) \rightarrow 3 \text{N}_2(g) + 2 \text{Na}(s)$. What volume of nitrogen gas, measured at 1.30 atm and 87°C , would be produced by the reaction of 70.0 g of NaN_3 ?

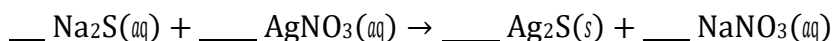
Molarity Stoichiometry: Solutions to All Your Problems

***Note:** millimoles (or mmol) can be your BEST friend during solution stoich!

$$\frac{\text{mmol}}{\text{mL}} = M$$

1. 250 mL of 0.70 M Li_3PO_4 and 250 mL of excess $\text{Ca}(\text{OH})_2$ are mixed, producing an aqueous LiOH solution (and a calcium phosphate precipitate). What is the molar concentration of LiOH in this solution?

2. Calculate the mass of Ag_2S produced when 125 mL of 0.200 M AgNO_3 is added to excess Na_2S solution.



3. 50.0 mL of $\text{Mg}(\text{OH})_2$ is used to neutralize 35.0 mL of 0.60 M HBr . What is the molarity of the $\text{Mg}(\text{OH})_2$?

Molarity Stoich Shortcut: $M_1V_1 = M_2V_2$

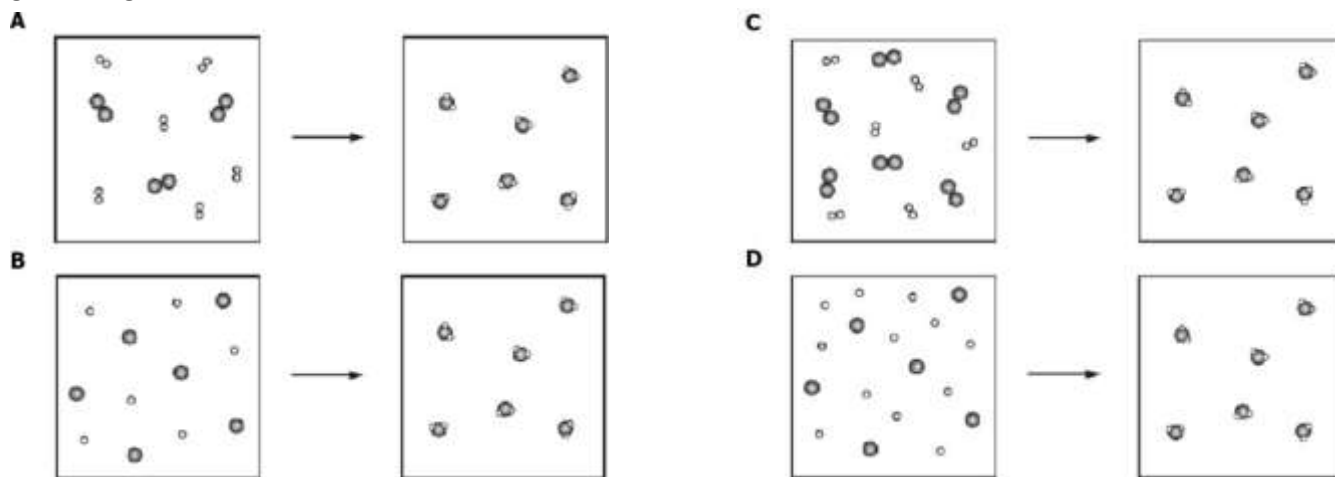
This shortcut is useful if:

- IF you are starting and ending with molarity, and
- the species compared have a 1:1 mole ratio

4. 125 mL of KOH completely neutralizes 57.0 mL of 1.20 M HBr . Calculate the initial concentration of KOH .

Let's Get Stoiched! Multiple Choice Practice

1. Which of the following particulate diagrams best shows the formation of water vapor from hydrogen gas and oxygen gas in a rigid container at 125°C?



2. A sample of 9.00 grams of aluminum metal is added to an excess of hydrochloric acid. The volume of hydrogen gas produced at standard temperature and pressure is:

a. 22.4 L b. 11.2 L c. 7.46 L d. 5.60 L

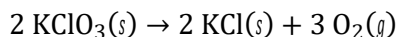
3. Upon combustion, a 30. g sample of a compound containing only carbon, hydrogen, and oxygen produces 44.0 g of carbon dioxide and 18.0 g of water. Find the empirical formula of the compound.

a. CHO b. CH₂O c. C₂HO₂ d. CH₂O₂

4. A student titrates 20.0 mL of 1.0 M NaOH with 2.0 M formic acid, HCO₂H ($K_a = 1.8 \times 10^{-4}$). Formic acid is a monoprotic acid (it only donates 1 H⁺ ion). How much formic acid is needed to completely neutralize the NaOH?

a. 10.0 mL b. 20.0 mL c. 30.0 mL d. 40.0 mL

Use the following information to answer #5-6: When heated in a closed container in the presence of a catalyst, 1.2×10^{23} formula units of potassium chlorate decompose into potassium chloride and oxygen gas via the following reaction:



5. How many grams of oxygen gas will be generated?

a. 1.60 g b. 4.80 g c. 9.60 g d. 18.37

6. Approximately how many liters of oxygen gas will be evolved at STP?

a. 1.24 L b. 3.36 L c. 6.72 L d. 22.4 L

Limiting and Excess Reagents

Limiting reactant: will be completely _____ during the chemical reaction → determines all other amount

Excess reactant: will _____ be completely used up during the chemical reaction → has some left over at the end

Percent Yield: a method to calculate the effectiveness of a chemical reaction.

- **Actual yield:** what you produce from actually doing the reaction in a _____ setting.
- **Theoretical yield:** what “should have been” produced from chemical reaction. This is calculated with _____!

$$\text{Percent Yield} = \frac{\text{Actual Yield}}{\text{Theoretical Yield}} \times 100$$

How to Find the Limiting Reactant: Two Methods

1. Calculate the number of _____ of each reactant and then divide each one by its _____ from the balanced reaction.

→ The reactant which produces the smaller number will be the limiting reactant.

Advantage: very fast (great for multiple choice!)

Disadvantage: you still have to actually calculate whatever info is asked for by the problem.

2. Use a RICE table to calculate the ending number of _____ for both reactants and products.

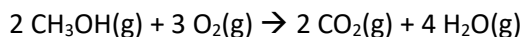
→ The reactant which is entirely used up will be the limiting reactant.

Advantage: you will calculate the ending amount of your excess reactant AND products, so any calculation you need is already complete.

Disadvantage: you have to be in MOLES, so you might have extra math to do before and after.

Let's try BOTH methods! Which do you like best?

1. Methanol is combusted according to the balanced equation below. If you have 50.0 g of methanol and 50.0 g of molecular oxygen, how many grams of excess reactant remain unreacted?

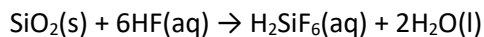


2. Aqueous sodium hydroxide reacts with phosphoric acid to give sodium phosphate and water in the following balanced chemical equation: $3 \text{NaOH (aq)} + \text{H}_3\text{PO}_4 \text{ (aq)} \rightarrow \text{Na}_3\text{PO}_4 \text{ (aq)} + 3 \text{H}_2\text{O (l)}$

If 20.80 g of NaOH is mixed with 29.40 g of H_3PO_4 ($\text{NaOH} = 40.00 \text{ g/mol}$, $\text{H}_3\text{PO}_4 = 98.00 \text{ g/mol}$):

- a. How many grams of Na_3PO_4 can be formed?
- b. How many grams of the excess reactant remain unreacted?
- c. If the actual yield of Na_3PO_4 was 15.00 g, what is the percent yield of Na_3PO_4 ?

3. Hydrofluoric acid solutions cannot be stored in glass containers because HF reacts readily with silica in glass to produce hexafluorosilicic acid (H_2SiF_6).



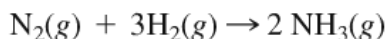
If 40.0 kg silicon dioxide and 40.0 kg of HF react:

- a. Determine the mass of the excess reactant remaining.
- b. Determine the theoretical yield of hexafluorosilicic acid.
- c. Determine the percent yield if the actual yield is 45.8 kg H_2SiF_6 .

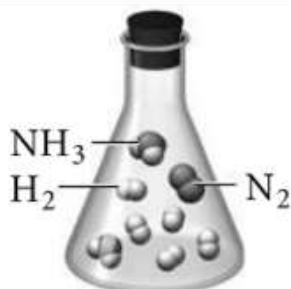
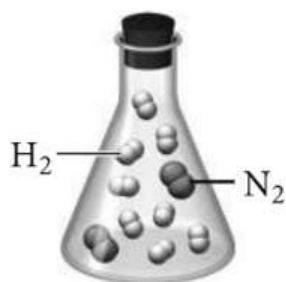
Multiple Choice Practice

4.

Nitrogen and hydrogen gas react to form ammonia according to the reaction:



If a flask contains a mixture of reactants represented by the image at right, which image below best represents the mixture in the flask after the reactants have reacted as completely as possible? What is the limiting reactant? Which reactant is in excess?



(a)

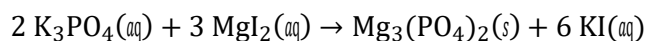


(b)

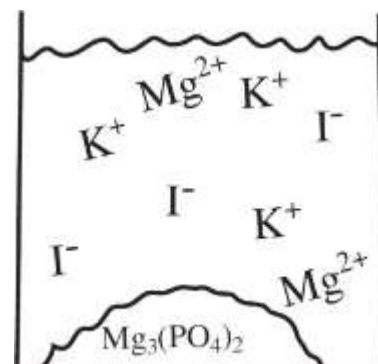


(c)

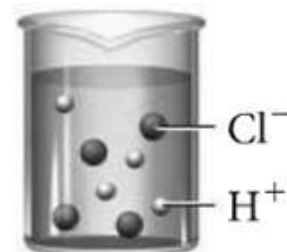
5. Based on the particulate drawing of the products for the reaction below, which reactant is limiting for the following reaction and why?



- The K_3PO_4 , because there are no excess PO_4^{3-} ions after the reaction.
- The MgI_2 , because there are excess Mg^{2+} cations remaining after the reaction.
- The K_3PO_4 , because it contains a cation that cannot form a precipitate.
- The MgI_2 , because it required more of itself to create the products.



6. A hydrochloric acid solution will neutralize a sodium hydroxide solution. Look at the molecular views showing one beaker of HCl and four beakers of NaOH. Which NaOH beaker will just neutralize the HCl beaker? Begin by writing a balanced chemical equation for the neutralization reaction.



(a)



(b)

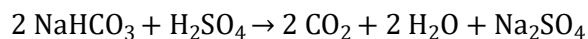


(c)



(d)

7. What is the maximum grams of CO_2 (molar mass 44.0 g/mol) that can be produced from 50.0 mol each of sulfuric acid and sodium hydrogen carbonate? The balanced equation is:



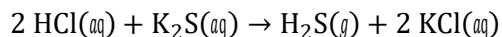
- a. (50.0) (1/2) (44.0) c. (50.0) (4) (44.0)
b. (50.0) (44.0) d. (50.0) (2) (44.0)

8. The reaction of silver metal and dilute nitric acid proceeds according to the equation below. If 0.10 mol of powdered silver is added to 10. mL of 6.0 molar nitric acid, the number of moles of NO gas that can be formed is:



- a. 0.015 mol b. 0.030 mol c. 0.045 mol d. 0.020 mol

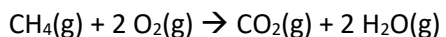
9. 250. mL of 0.30 M hydrochloric acid is combined with 250. mL of 0.10 M K_2S in the gas evolution shown below.



What is the concentration of K^+ ions in the KCl solution formed during this reaction?

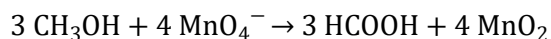
- a. 0.050 M b. 0.10 M c. 0.15 M d. 0.30 M

10. Which of the following statements about the reaction below is **false**?



- a. Every methane molecule that reacts produces two water molecules.
- b. If 16.0 g of methane react with 64.0 g of molecular oxygen, the combined masses of the products will be 80.0 g.
- c. If 11.2 L of methane react with an excess of molecular oxygen, the volume of CO_2 produced at STP is $(44/16)(11.2)$ liters.
- d. If 22.4 L of methane at STP react with 64.0 g of molecular oxygen, 22.4 L of CO_2 at STP can be produced.

11. When 16 g of methanol (CH_3OH) was treated with excess oxidizing agent (MnO_4^-), 16 g of formic acid (HCOOH) was obtained. Using the following chemical equation, calculate the percent yield.



- a. 100% b. 90% c. 80% d. 70%

12. How many grams of NH_3 can be prepared from 77.3 g of N_2 and 14.2 g of H_2 ?

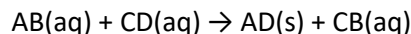
- a. 47.0 g b. 79.7 g c. 93.9 g d. 120. g

Types of Chemical Reactions

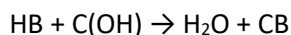
Type of Reaction	What Happens	Common Form
Synthesis Reaction	Two or more substances come together to make _____ new substance.	$A + B \rightarrow C$
Decomposition Reaction	One substance breaks down into two or more simpler substances.	$C \rightarrow A + B$
Single Replacement Reaction	One element replaces another in a compound.	$A + BC \rightarrow AC + B$
Double Replacement Reaction	Ions of two compounds exchange places to form two new compounds.	$AB + CD \rightarrow AD + CB$
Combustion reaction	A substance combines with _____ gas, releasing large amounts of energy in the form of heat and light.	$C_xH_yO_z + O_2 \rightarrow H_2O + CO_2$

There are three subsets of double replacement reactions that you will need to know!

1. Precipitation Reaction: two aqueous solutions mix to form a precipitate (_____) $\rightarrow$ more about this shortly!



2. Neutralization (Acid-Base) Reaction: an Arrhenius acid and base react to produce a salt and water.



3. Gas Evolution Reaction: two aqueous solutions mix to form a gas which _____ out of solution. There are four general categories of this reaction type!

Compounds that Undergo Gas-Evolution Reactions

Reactant Type	Intermediate Product	Gas evolved (with decomposed products, if applicable)
Sulfides	<i>None</i>	$H_2S(g)$
Carbonates and bicarbonates	H_2CO_3	$CO_2(g) + H_2O(l)$
Sulfites and bisulfites	H_2SO_3	$SO_2(g) + H_2O(l)$
Ammonium	NH_4OH	$NH_3(g) + H_2O(l)$

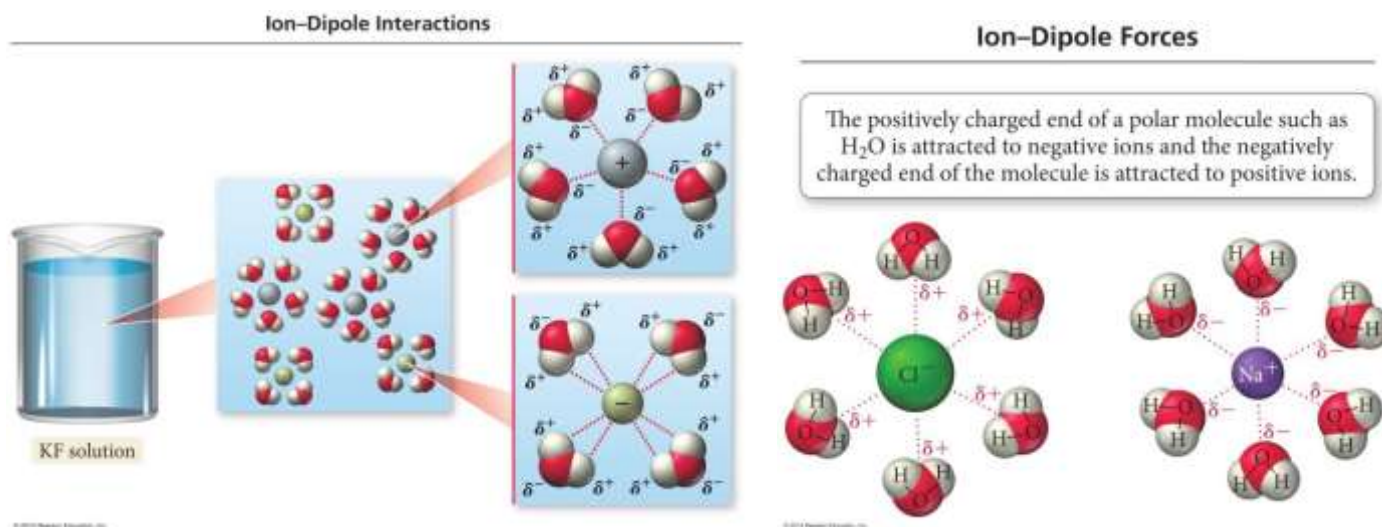
Precipitation Reactions

Precipitation Reactions: a double replacement reaction between aqueous solutions of ionic compounds which produces

an ionic compound that is _____ in water: this insoluble product is called a precipitate.

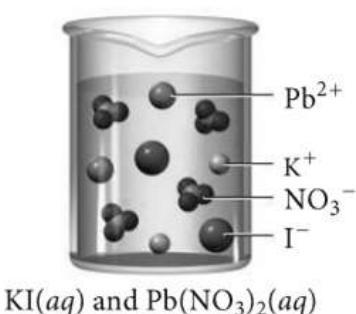
Quick review: Ionic Compound Solubility

- When we mix a solute with a solvent, there are attractive forces (ion-dipole IMFs) between the solute and solvent particles – if the attraction is strong enough, this is what allows the solute to _____!
- Ions separate (ionize) from one another when dissolved in water (called _____)
- The number of ions produced in solution depends on the ratio in the original _____.
 - Ex: $\text{Pb}(\text{NO}_3)_2$ dissociates to form $\rightarrow$
 - Thus, **1 formula unit** of $\text{Pb}(\text{NO}_3)_2$ dissociates to form _____ total ions.

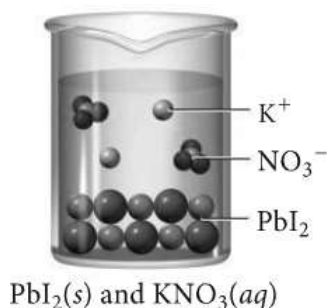


Back to precipitation reactions!

For example, consider what happens if you mix two solutions, KI(aq) and $\text{Pb(NO}_3)_2\text{(aq)}$. Before mixing, both compounds are _____ in water: they have **dissociated into ions** in their respective solutions. When the two solutions are mixed together:



In the *instant* that the solutions come into contact, all four ions are present:



In this reaction, KNO_3 is soluble, but PbI_2 is insoluble. Thus, PbI_2 precipitates.

Solubility Rules

There are many rules to determine which ions are soluble in water. Although you no longer need to know the very large list of solubility rules from the AP test of yore, there are some basic solubility rules you **MUST** know!

Must be memorized: _____ **metal cations, NH_4^+ , $\text{C}_2\text{H}_3\text{O}_2^-$, and NO_3^- are always soluble**

Other rules (will be given within a problem or can be figured out by context- Name a context clue: _____):

- Always soluble: ClO_3^- , ClO_4^- , HCO_3^-
- Generally soluble:
 - Bromide, chloride, and iodide anions are soluble except when combined with Ag^+ , Pb^{2+} , and Hg_2^{2+} .
 - SO_4^{2-} is soluble except when combined with Sr^{2+} , Ba^{2+} , Pb^{2+} , and Hg_2^{2+} .
- Generally insoluble:
 - OH^- and S^{2-} are insoluble except when combined with Ca^{2+} , Sr^{2+} , Ba^{2+} , (and things from rule 1).
 - CO_3^{2-} , PO_4^{3-} , SO_3^{2-} , and CrO_4^{2-} are insoluble except when combined with things from rule 1.

Note: Precipitation reactions do not always occur when two aqueous solutions are mixed!

	Precipitation Rxn?	Why?
$\text{RbCl(aq)} + \text{NaBr(aq)} \rightarrow \text{RbBr(aq)} + \text{NaCl(aq)}$	Nope.	No insoluble substance produced.
$\text{HCl(aq)} + \text{AgNO}_3\text{(aq)} \rightarrow \text{AgCl(s)} + \text{HNO}_3\text{(aq)}$	Yes!	Solid produced from aqueous reactants.
$3 \text{ MnO}_2\text{(aq)} + 4 \text{ Al(s)} \rightarrow 3 \text{ Mn(s)} + 2 \text{ Al}_2\text{O}_3\text{(aq)}$	Nope.	Not starting from only aqueous reactants.

Ways to Represent a Precipitation Reaction

Complete ionic equation: Starting from the full equation, split any aqueous substances into their ions, but leave non-aqueous substances intact.

Net ionic equation: Starting from the complete ionic equation, remove all ions that found in exactly the same form on both sides of the equation (called spectator ions, since they do not participate in reaction).

Example

Molecular (Full) Equation: $2\text{AgNO}_3(\text{aq}) + \text{CaCl}_2(\text{aq}) \rightarrow 2\text{AgCl}(\text{s}) + \text{Ca}(\text{NO}_3)_2(\text{aq})$

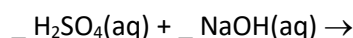
Complete Ionic Equation: $2\text{Ag}^+(\text{aq}) + 2\text{NO}_3^-(\text{aq}) + \text{Ca}^{2+}(\text{aq}) + 2\text{Cl}^-(\text{aq}) \rightarrow 2\text{AgCl}(\text{s}) + \text{Ca}^{2+}(\text{aq}) + 2\text{NO}_3^-(\text{aq})$

Net ionic equation: $2\text{Ag}^+(\text{aq}) + 2\text{Cl}^-(\text{aq}) \rightarrow 2\text{AgCl}(\text{s})$ simplifies to:

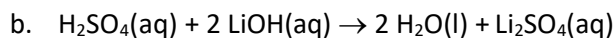
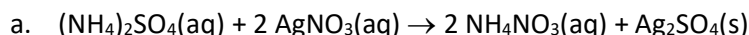
Let's Try!

1. Why is a precipitation reaction best represented by a net ionic equation, rather than the complete or molecular equation?

2. Complete and balance the following reaction, then write the complete AND net ionic equations.



3. Write the net ionic equation for the following reactions.

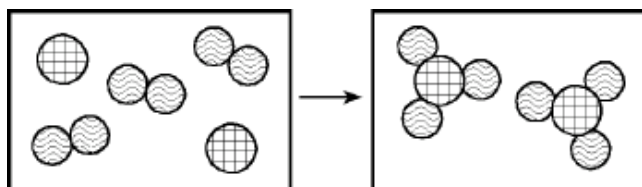


Don't Overreact: More Reactions Practice!

Part I: Complete the following chart. (Balance the reactions if you want more practice!)

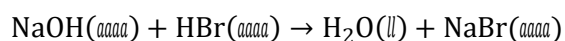
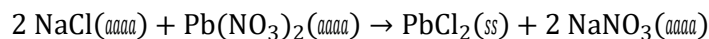
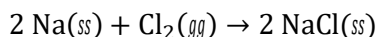
Reaction	Type of Reaction A/B, PPT, Redox?
1. $\text{Ca(s)} + \text{H}_2\text{O(l)} \rightarrow \text{Ca(OH)}_2\text{(aq)} + \text{H}_2\text{(g)}$	
2. $\text{NH}_4\text{F(aq)} + \text{Sr(ClO}_3)_2 \rightarrow \text{NH}_4\text{ClO}_3\text{(aq)} + \text{SrF}_2\text{(aq)}$	
3. $\text{HNO}_3\text{(aq)} + \text{LiOH(aq)} \rightarrow \text{LiNO}_3\text{(aq)} + \text{H}_2\text{O(l)}$	
4. $\text{LiI(aq)} + \text{Pb(NO}_3)_2\text{(aq)} \rightarrow \text{LiNO}_3\text{(aq)} + \text{PbI}_2\text{(s)}$	
5. $\text{C}_6\text{H}_{14}\text{(g)} + \text{O}_2\text{(g)} \rightarrow \text{H}_2\text{O(l)} + \text{CO}_2\text{(g)}$	

Part II: Multiple Choice Practice



- The diagram above best represents which type of reaction?
 - Acid/base
 - Oxidation/reduction
 - Precipitation
 - Decomposition
- If we dissolve 25 grams of salt in 251 grams of water, what is the mass of the resulting solution?
 - 251 g
 - 276 g
 - 226 g

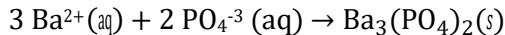
3. Consider the following three equations for chemical reactions:



These are examples of:

- three redox reactions
- three acid-base reactions
- a redox reaction, a precipitation reaction, and an acid-base reaction
- a neutralization reaction, then two precipitation reactions

4. Given the net ionic equation



how many grams of iron (II) phosphate must be present to react with 2.0×10^2 grams of barium bromide?

(MW $\text{Fe}_3(\text{PO}_4)_2 = 357 \text{ g/mol}$, MW $\text{BaBr}_2 = 297 \text{ g/mol}$)

A

$$200 \text{ g BaBr}_2 \times \frac{1 \text{ mol BaBr}_2}{297 \text{ g BaBr}_2} \times \frac{1 \text{ mol Fe}_3(\text{PO}_4)_2}{3 \text{ mol BaBr}_2} \times \frac{357 \text{ g Fe}_3(\text{PO}_4)_2}{1 \text{ mol Fe}_3(\text{PO}_4)_2}$$

B

$$200 \text{ g BaBr}_2 \times \frac{1 \text{ mol BaBr}_2}{297 \text{ g BaBr}_2} \times \frac{2 \text{ mol Fe}_3(\text{PO}_4)_2}{3 \text{ mol BaBr}_2} \times \frac{357 \text{ g Fe}_3(\text{PO}_4)_2}{1 \text{ mol Fe}_3(\text{PO}_4)_2}$$

C

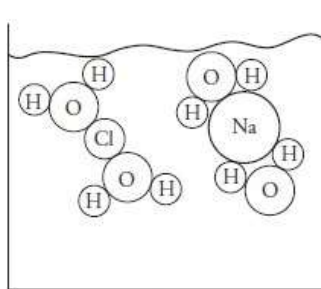
$$200 \text{ g BaBr}_2 \times \frac{1 \text{ mol BaBr}_2}{297 \text{ g BaBr}_2} \times \frac{3 \text{ mol Fe}_3(\text{PO}_4)_2}{1 \text{ mol BaBr}_2} \times \frac{357 \text{ g Fe}_3(\text{PO}_4)_2}{1 \text{ mol Fe}_3(\text{PO}_4)_2}$$

D

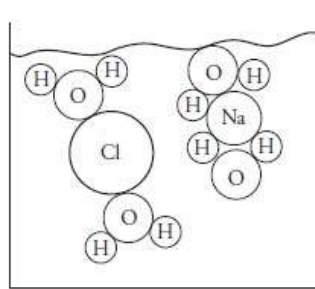
$$200 \text{ g BaBr}_2 \times \frac{1 \text{ mol BaBr}_2}{297 \text{ g BaBr}_2} \times \frac{3 \text{ mol Fe}_3(\text{PO}_4)_2}{2 \text{ mol BaBr}_2} \times \frac{357 \text{ g Fe}_3(\text{PO}_4)_2}{1 \text{ mol Fe}_3(\text{PO}_4)_2}$$

5. If solutions containing equimolar amounts of AgNO_3 and KCl are mixed, what is the identity of the spectator ions?
- Ag^+ , NO_3^- , K^+ , and Cl^-
 - Ag^+ and Cl^-
 - Ag^+ and K^+
 - NO_3^- and K^+
6. Choose the correct net ionic equation representing the reaction that occurs when solutions of potassium carbonate and copper (I) chloride are mixed.
- $\text{K}_2\text{CO}_3(\text{aq}) + 2 \text{CuCl}(\text{aq}) \rightarrow 2 \text{KCl}(\text{aq}) + \text{Cu}_2\text{CO}_3(\text{s})$
 - $\text{K}_2\text{CO}_3(\text{aq}) + 2 \text{CuCl}(\text{aq}) \rightarrow 2 \text{KCl}(\text{s}) + \text{Cu}_2\text{CO}_3(\text{aq})$
 - $\text{CO}_3^{2-}(\text{aq}) + 2 \text{Cu}^+(\text{aq}) \rightarrow \text{Cu}_2\text{CO}_3(\text{s})$
 - $\text{K}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{KCl}(\text{s})$

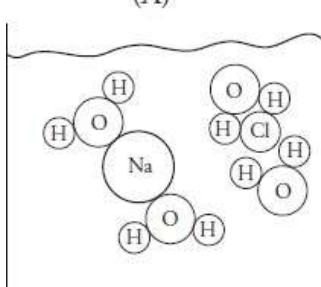
7. Which of the following diagrams best represents what is happening on a molecular level when NaCl dissolves in water?



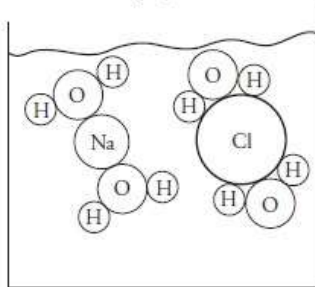
(A)



(C)



(B)



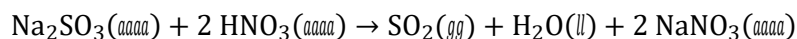
(D)

Use this information to answer #8-9: When sodium sulfite(aq) is added to nitric acid, aqueous sodium nitrate and two other products are formed.

8. What is the balanced net-ionic equation for this reaction?

- | | |
|--|---|
| a. $\text{Na}_2\text{SO}_3(\text{aq}) + 2 \text{HNO}_3(\text{aq}) \rightarrow 2 \text{NaNO}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{SO}_2(\text{g})$ | c. $2 \text{H}^+(\text{aq}) + \text{SO}_3^{2-}(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l}) + \text{SO}_2(\text{g})$ |
| b. $\text{Na}_2\text{SO}_3(\text{aq}) + 2 \text{HNO}_3(\text{aq}) \rightarrow 2 \text{NaNO}_3(\text{aq}) + \text{H}_2\text{SO}_3(\text{aq})$ | d. $2 \text{H}^+(\text{aq}) + \text{SO}_3^{2-}(\text{aq}) \rightarrow \text{H}_2\text{SO}_3(\text{aq})$ |

9. If 0.62 g of sodium sulfite is added to excess nitric acid, how many mL of SO_2 gas will be evolved at 625 mmHg and 27°C ? (The molar mass of sodium sulfite is 126 g/mol).



- | | | | |
|-------------|------------|----------|-----------|
| a. 0.018 mL | b. 0.20 mL | c. 13 mL | d. 150 mL |
|-------------|------------|----------|-----------|

Stoichiometry in the Lab: Titrations and Gravimetric Analysis!

Titration (Volumetric Analysis): a laboratory procedure in which a substance in a solution of known concentration (the titrant, usually in a buret) is reacted with another substance in a solution of unknown concentration (the analyte) in order to determine the unknown concentration. Often used with neutralization reactions!

- **Equivalence point:** the point in a titration when the added solute (titrant) completely reacts with the analyte. In an acid-base titration, the point at which the amount of acid is *stoichiometrically equal* to the amount of base in solution.

At equivalence point, $\text{moles}_{\text{H}^+} = \text{moles}_{\text{OH}^-}$

- **End point:** the point of pH change where an indicator changes color (if thoughtfully chosen, this occurs approximately at the equivalence point).

Example: It is found that 24.68 mL of 0.1165 M NaOH is needed to titrate 0.2931 g of an unknown monoprotic acid to the phenolphthalein end point.

a. How many moles of acid were present in the sample?

b. Calculate the molar mass of the unknown acid.

Gravimetric Analysis – a laboratory procedure in which an ion is precipitated out of a mixture in order to find the percent mass of that ion in an impure substance.

Method:

1. Reactant impure material with a known compound, forming a _____ which contains the ion of interest.
2. Filter and dry precipitate, then measure its mass.
3. Use stoichiometry calculations to determine the _____ of the ion of interest, using the balanced reaction to work backwards from the mass of the precipitate measured in the lab.

Steps in Gravimetric Analysis (Lab Technique)

1. Dry and weigh impure sample.
2. _____ sample.
3. Add precipitating reagent in _____.
4. Filter precipitate from solution.
5. _____ precipitate to remove impurities.
6. Dry and weigh to constant _____ (± 0.03 g)



Steps in Gravimetric Analysis

1. Write the balanced precipitation reaction.
2. Use stoichiometry to convert from the mass of your precipitate to the mass of the ion of interest.

3. Calculate % by mass: $\% \text{ mass of ion} = \frac{\text{mass of ion (calculated in step 2)}}{\text{mass of original sample}} \times 100$

Example: The following data were obtained when a sample of unknown chloride was analyzed:

Mass of watch glass	8.2030 g
Mass of unknown chloride sample + watch glass	8.3014 g
Mass of AgCl(s) + watch glass (after drying)	8.432 g

- a. Calculate the mass of the precipitate.
- b. Calculate the mass of chloride ions in the original sample.
- c. Calculate the percent mass of chloride ions in the sample.

Common Lab Errors in Gravimetric Analysis

Error	Effect on Calculated % by Mass of Ion
Failure to add precipitating reagent in excess.	- Not all of the ion of interest will precipitate, thus some of the ion will be lost in the filtrate. - The calculated ion % will be _____ than the actual ion % by mass.
Failure to wash the precipitate to remove soluble impurities.	- Some impurities will be left in the precipitate, so measured mass includes both the precipitate and these contaminants. - The calculated ion % will be _____ than the actual ion % by mass.
The precipitate is excessively washed or washed with warm water.	- Although the precipitate is "insoluble", small amounts of precipitate will dissolve into the wash water and be carried into the filtrate. - The calculated ion % will be _____ than the actual ion % by mass.
The precipitate is not dried to "constant mass".	- Not all of the water molecules will have been removed, so measured mass includes both the precipitate and remaining water molecules. - The calculated ion % will be _____ than the actual ion % by mass.

Let's practice!

1. A 3.187 g impure sample containing sulfur and an inert material was to be analyzed for sulfur. As part of the procedure, the sample is dissolved in 350. mL of water, and the sulfur is converted to sulfate ion, SO_4^{2-} . Barium nitrate is added, which causes the sulfate to precipitate out as BaSO_4 . The dried BaSO_4 has a mass of 2.005 g.
 - a. What mass of sulfur was in the original sample?
 - b. What is the percent of sulfur in the original ore?
 - c. If the inert material was potassium carbonate, what is the molar ratio of sulfur to potassium carbonate in the original sample?
 - d. If the original sample was instead dissolved into 500. mL of water, what effect would this have on the calculated percent by mass of sulfur?
2. Calculate the molarity of a $\text{Sr}(\text{OH})_2$ solution if 32.5 mL of it is needed to neutralize 20.88 mL of 3.50 M HClO_4 .
3. Calculate the volume of a 5.0 M solution of HNO_3 needed to neutralize 6.2 g of NaOH .
4. You wish to determine the molarity of a solution of sodium hydroxide. To do this, you titrate a 25.00 mL aliquot of your sample (which has 3 drops of added phenolphthalein indicator so that it is pink) with 0.1067 M HCl . The sample turns clear (indicating that the $\text{NaOH}(\text{aq})$ has been precisely neutralized by the HCl solution) after the addition of 42.95 mL of the HCl .
 - a. Calculate the molarity of your NaOH solution.
 - b. If your lab partner added excess HCl to the solution, passing the equivalence point, what would be the effect on the calculated molarity of your NaOH solution?

Multiple Choice Practice:

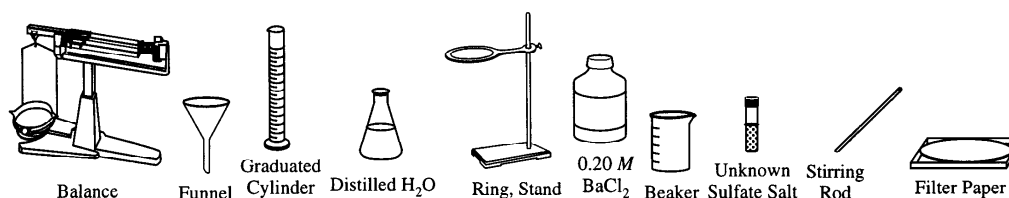
1. A sample of an unknown chloride compound was dissolved in water, and then titrated with excess $\text{Pb}(\text{NO}_3)_2$ to create a precipitate. After drying, it is determined there are 0.0050 mol of precipitate present. What mass of chloride is present in the original sample?
a. 0.177 g b. 0.355 g c. 0.522 g d. 0.710 g
2. Gallic acid is known to contain carbon, hydrogen, and oxygen. A chemist wanting to determine the molecular formula of gallic acid burned 85 g of the compound in an elemental analyzer. The products of the combustion were 154 g of CO_2 and 27 g of H_2O . Which of the following could be the formula of the compound?
a. $\text{C}_7\text{H}_6\text{O}_5$ b. $\text{C}_{11}\text{H}_7\text{O}$ c. $\text{C}_3\text{H}_8\text{O}_2$ d. $\text{C}_9\text{H}_{12}\text{O}$
3. 200 mL of 1.5 M HCl is mixed with 150 mL of 3.0 M NaOH. What is the molarity of the resulting salt solution?
a. 0.45 M NaCl b. 2.0 M NaCl c. 0.86 M NaCl d. 1.5 M NaCl
4. How many milliliters of 1.00 M HCl will completely neutralize 100 milliliters of 1.00 M NaOH?
a. 100.0 mL b. 200.0 mL c. 50.0 mL d. 1.00 mL
5. 0.60 M HNO_3 was used to neutralize 15 mL of 0.30 M $\text{Sr}(\text{OH})_2$. What volume of HNO_3 was needed?
a. 7.5 mL b. 15.0 mL c. 22.5 mL d. 30.0 mL
6. The complete neutralization of 15.0 mL of KOH requires 0.025 mol H_2SO_4 . The [KOH] was:
a. 1.50 M b. 1.67 M c. 3.33 M d. 6.67 M

Unit 2: AP Quiz Free Response Practice [2003 Form B #2, modified, 6 points]

1. Solid iron (III) oxide can be reduced with gaseous carbon monoxide, producing solid iron and carbon dioxide gas.
 - a. In an experiment, a student combines a 16.2 L sample of CO(g) at 1.50 atm and $200.^{\circ}\text{C}$ with 15.39 g of $\text{Fe}_2\text{O}_3\text{(s)}$.
 - i. What is the balanced chemical equation for this reaction? [2 points]
 - ii. How many moles of CO(g) are available for the reaction? [1 point]
 - iii. What is the limiting reactant for the reaction? Justify your answer with calculations. [2 points]
 - iv. How many moles of Fe(s) are formed in the reaction? [1 point]

Unit 2: AP Free Response Practice #1 [6 points]

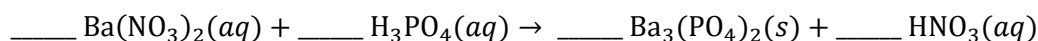
2. A 3.00 g sample of impure solid, containing sodium chloride and an inert substance, is dissolved in 200. mL of water. An excess of aqueous silver ions is added, and reaction occurs, producing a white precipitate of silver chloride. After filtering and drying, it is found that 0.450 g of solid is produced.
 - a. Write the balanced equation for the reaction which occurs, including states. [2 points]
 - b. Write the balanced net ionic equation for the precipitation reaction, including states. [1 point]
 - c. Calculate the mass of chloride ions in the original impure sample. [2 points]
 - d. What percent by mass of the original sample was due to chloride ions? [1 point]

Unit 2: AP Free Response Practice #2 [8 points]

1. A student is performing an experiment to determine the mass percent of sulfate in an unknown soluble sulfate salt. The equipment shown above is available for the experiment. A drying oven is also available. Note: barium sulfate is insoluble in water.
 - a. Briefly list the steps needed to carry out this experiment. [2 points]
 - b. What experimental data needs to be collected to calculate the mass percent of sulfate in the unknown? [2 points]
 - c. Starting with an initial sulfate salt sample of 8.52 g, the student is able to recover 0.752 g of barium sulfate. What is the mass percent of sulfate in the original sample? [3 points]
 - d. If the student fails to dry the barium sulfate sample to constant mass, would this cause the calculated mass percent of sulfate to be less than, equal to, or greater than the actual mass percent? Justify your answer. [1 point]

Unit 2: AP Free Response Practice #3 [2003 Form B #2, modified, 5 points]

3. In a reaction vessel, 0.600 mol of Ba(NO₃)₂(s) and 0.300 mol of H₃PO₄(aq) are combined with deionized water to a final volume of 2.00 L. The reaction represented below occurs.



- a. Balance the equation above by writing the correct coefficients in the blanks provided. [1 point]
- b. Calculate the mass of Ba₃(PO₄)₂(s) formed. [2 points]
- c. Calculate the moles of nitric acid formed. [1 point]
- d. How many moles of excess reactant are left over once the reaction stops? [1 point]

Unit 2: AP Free Response Practice #4 [10 points]

3. 2.54 g of beryllium chloride are completely dissolved into 50.00 mL of water inside a beaker.
- a) Draw a particulate representation of all species in the beaker after the solute has dissolved. Your diagram should include at least one formula unit of beryllium chloride and four water molecules. Make sure the atoms and ions are correctly sized and oriented relative to each other. [2 points]



- b) Calculate the concentration of beryllium ions, chloride ions, and total ions in the beaker. [2 points]

A solution of 0.850 M lead (II) nitrate is then titrated into the beaker, causing a precipitate of lead (II) chloride to form.

- c) Identify the net ionic reaction occurring in the beaker. [1 point]
- d) Determine the volume of lead (II) nitrate in milliliters that must be added to the beaker to cause the maximum precipitate to form. [2 points]
- e) What is the theoretical yield of precipitate? [2 points]
- f) A student performing this lab has a percent error of 7.45%. They claim that the source of their error was dissolving the beryllium chloride in only 45.00 mL of water. Do you agree with the student? Justify your response. [1 point]

Unit 2: AP Free Response Practice #5 (2008 FR #3, modified) [10 points]

4. A student adds 150. mL of a 0.030 M lead (II) nitrate solution to 125 mL of 0.100 M sodium iodide solution.

a. A chemical reaction takes place and a precipitate is formed.

i. Write the balanced equation for the reaction, including states. [2 point]

ii. Write the balanced net ionic equation for the reaction, including states. [1 point]

b. Calculate the number of moles of each reactant. [2 points]

c. Identify the limiting reactant. Show calculations to support your identification. [1 point]

d. Calculate the molar concentration of $\text{NO}_3^-(\text{aq})$ in the mixture after the reaction is complete. [2 points]

e. Circle the diagram below that best represents the results after the mixture reacts as completely as possible. Explain the reasoning used in making your choice. [2 points]

